

Dot products

- dot product of vectors a and b : $a^T b = \sum a_i b_i$
- norm of x : $\|x\| = \sqrt{x^T x}$
- orthogonality if $a^T b = 0$
- when $Ax = 0$ means x is perpendicular to every row of A .
 - ↳ row 1 $\cdot x = 0$, row 2 $\cdot x = 0$...

Gram matrix $A^T A$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \quad A^T A = \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$$

- Gram is always symmetric: $G_{ij} = G_{ji}$ for $i \neq j$
- $A^T A$ is invertible exactly when has independent columns.

Covariance matrix:

1. Demean $A^T A$ columns to get $\tilde{R}$
2. Then $\Sigma = \tilde{R}^T \tilde{R} / (T-1)$ (T is number of observations)

$$\begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix} \xrightarrow{A^T A} \begin{bmatrix} 2.5 & 0 \\ -2.5 & 0 \end{bmatrix} \xrightarrow{\tilde{R}}$$

$$\begin{bmatrix} 2.5 & -2.5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2.5 & 0 \\ -2.5 & 0 \end{bmatrix} / (2-1) = \begin{bmatrix} 13 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{\tilde{R}^T \tilde{R} / (T-1)}$$

in Σ , Σ_{ii} is a variance,
 Σ_{ij} is a covariance.

correlation $\rho = \tilde{a}^T \tilde{b} / (\|\tilde{a}\| \|\tilde{b}\|) \in [-1, 1]$

portfolio variance $w^T \Sigma w$ and rank $\Sigma \leq \min(T-1, N)$